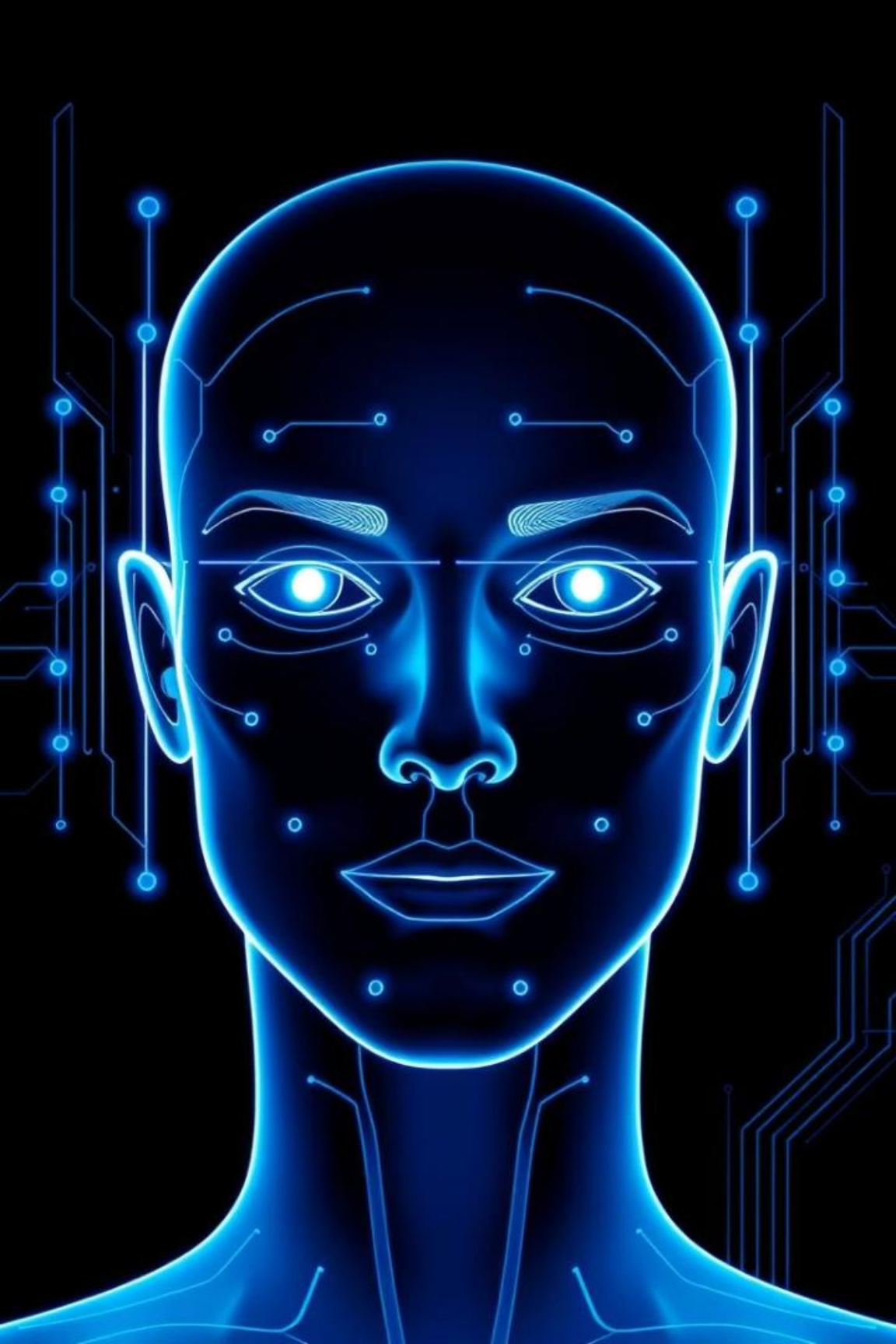




Face Recognition Attendance System

Automating Attendance Tracking with AI

Team:

Aditya Pandey 102303968

Yash Vardhan Singh 102303725

Mehul Kujur 102303712

Thapar Institute Of Engineering and Technology

Problem Statement

Manual System Issues

- Time-consuming roll calls delay classes
- Proxy attendance by friends
- Human errors and lost records

Our AI Solution

- AI-based facial recognition
- Zero physical contact, post-pandemic safe

Key Objectives

Automate Attendance

Real-time face detection replaces paper

High Accuracy

Over 85% recognition rate with LBPH

Prevent Fraud

Biometric verification stops proxy attendance

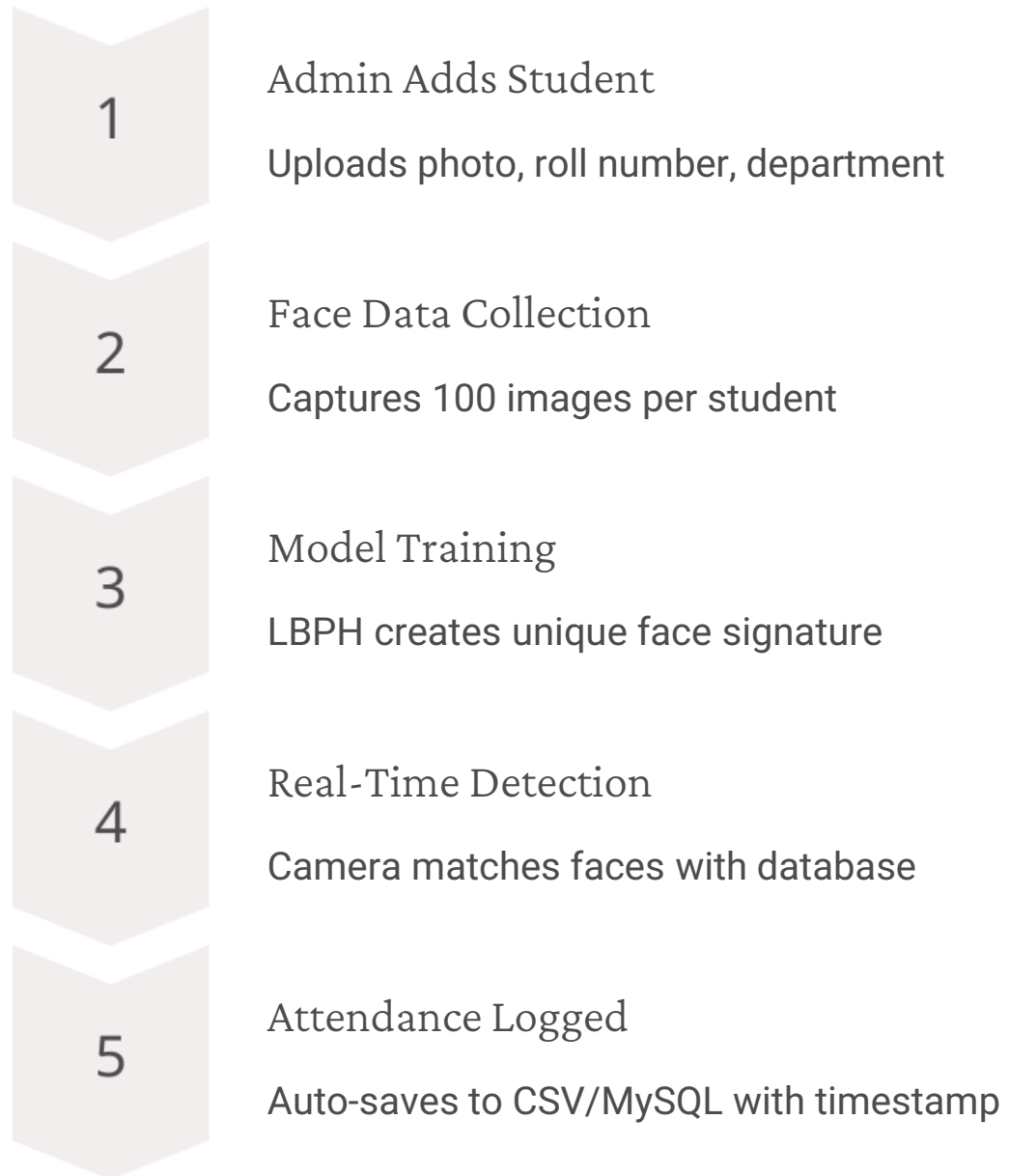
User-Friendly Interface

Simple GUI to add students and view logs

Scalability

Supports 1000+ students with MySQL

System Workflow



Technical Stack

Face Detection	Face Recognition	Database	GUI
OpenCV Haar Cascade	LBPH Algorithm: lightweight, accurate	MySQL stores roll numbers, departments	Tkinter, Python-compatible
Backend			
Python + OpenCV			

```

<
Farestnt_ont/// >
sauthor-parcasss/foresises //astratitier<
counper astA/betestSlle(Coler > wichefortrych; Destlors <
  cale0: { tuser<
    fapesspcracce({FPTRoneE(lplles > Tis wasst_gnstrise)
    {
      vole-astect/0presistes/fersatcotil/s cincturiam declab/;
      polesdrvol. (/vilestalt s <
        The-dofter,serte/E(Fblle contactief = 1.le)
      concat aplany a()
      checkg/luotronstes(colfe-factorciof erecolians-tistech/);
      The-duoP(atronst-cullife-faccutil/s cesoplan Dectob/)

      fesse-tance {oraur witchesl,rela
      fetpfiga({Freerantifersletorclcf trentilam meelaber
        crebase anlon/) face>
        inecnsttante loulle:,sectocid(Dectortring Dellep(inter '00.lar.r0f),
        intetathor = dutalle repteccliff tnsiplast.to.corny ~7);
    }
  }
}

<
Poutcalegsedfovtorgelle(iontassftudice!: (/plaler))
ferecesrratce({FTTrosFALITE)) sprto/fs tsop/foun_netactions
chotaassfattutto (/Lettied.1s te.coles
  fhocraysice({Fbeletcalte-ratattcl0: varoptiim deptliect/as
    <derag agat = lateer
  soterchere/ligetiffcoptedll0-/SS(1oTow,P5);
  cheetevior = ((intercian.cotleel10CPelefl; 307);
  drackg/iace(FFPiThonalle-ractectl0SBalcacia.F5);
crlestriedl; APP
  face-deteacion -cutratiferrattoralIt (hiet, =49,icd.-11,esl =));

rauthor:/Faciler: tcyisters((
  fne-canter({Frelerleterst/iaconcler ( Ins'Uorch <10.led <
    matestarler = VuillER raptertliif capritive csyttor Opless/1llsvesfigm at., =({
    Dntecty.Sotecr-culled tnpayctar.alass.2199);

    inerstanter = cytalen replialle Wregpoller enpacturser =({; dof1)
    intesstoabllam vartert:= mulluag (0ETB));
    inecrstater = cyollle-iptioel = Uppriciles .ervelirer =({).5#5);
    brnatsete/Factter= Datay rentact:
      materlog/aettation= itb-acp/unarley" T:SL octu15)),
      eapern-(LRPR)
      datession = Dviller "/rwicnl_ie0.41;

```

Code Highlights

Face Detection

detectMultiScale for real-time face detection

Database Integration

Insert student data into MySQL

Attendance Logging

Append attendance with timestamp to CSV

Dataset & Training

Dataset Structure

Folder: data/ with user.{rollno}.{id}.jpg files

Training Process

- Convert images to grayscale
- Extract LBPH features
- Save model as classifier.xml

Results & Performance

- Accuracy: 85-90% in good lighting
- Speed: 10-15 faces/second at 720p
- Storage: MySQL handles 10,000+ records
- CSV for daily backups

Metric	Traditional	Our System
Time/Student	30 sec	2 sec
Proxy Risk	High	None

Challenges & Fixes

Challenge	Solution
Low-light recognition	Adaptive thresholding added
False positives	Increased training samples
Database lag	Batch processing implemented



Test Metrics

Face Detection Report:

- Precision: 0.89
- Recall: 0.92
- F1-Score: 0.90

LBPH Recognition Accuracy: 87.5%

FPS: 12.3

Noise Robustness: 8% drop
(Gaussian)

Database Query: 0.32 sec

Future Scope

- 1 — 2024: Mobile App
Android/iOS attendance marking
- 2 — 2025: Cloud Sync
Live logs on AWS/Azure
- 3 — Deep Learning Upgrade
CNN for 95%+ accuracy
- 4 — Multi-Factor Auth
Face + RFID/NFC

